Extra Dessert #13: Station Restrictions and Special Operations

Spurious emissions, FCC monitoring stations, National Radio Quiet Zone, antenna structure rules, PRB-1, RACES.

Group E1B · 11 pool questions · Self-study reference · Practice exams at eham.net/exams

Spurious Emissions And Bandwidth

Question	Answer	Q
What is a spurious emission?	**An emission outside the signal's necessary bandwidth that can be reduced or eliminated without affecting the information transmitted** (E1B01).	E1B01
Acceptable bandwidth for digital voice or SSTV on HF	**3 kHz** (E1B02).	E1B02
If amateur signal interferes with a good-engineering broadcast receiver, the FCC may	**Require the amateur to avoid transmitting during certain hours on the offending frequencies** (E1B08). Cannot simply shut you down.	E1B08

Protected Locations

Question	Answer	Q
Distance to protect an FCC monitoring facility	**1 mile** (E1B03).	E1B03
What is the National Radio Quiet Zone?	**An area surrounding the National Radio Astronomy Observatory** (E1B05) – in West Virginia.	E1B05
70 cm repeater interferes with radiolocation – what then?	**Cease operation or make changes to mitigate the interference** (E1B04). The amateur is secondary on 70 cm.	E1B04

Antenna Structures And PRB-1

Question	Answer	Q
Erecting antenna near a public-use airport	**You may have to notify the FAA and register with the FCC under Part 17** (E1B06).	E1B06
PRB-1 applies to what type of regulations?	**State and local zoning** (E1B07). Federal preemption protecting amateur antenna structures.	E1B07
What does PRB-1 require?	**Reasonable accommodations of amateur radio must be made** (E1B11). Not "no limits", but state/local regs cannot effectively ban ham antennas.	E1B11

RACES (Radio Amateur Civil Emergency Service)

Question	Answer	Q
Which amateur stations may operate under RACES?	**Any FCC-licensed amateur station certified by the responsible civil defense organization for the area served** (E1B09).	E1B09
Frequencies authorized to a RACES station	**All amateur service frequencies authorized to the control operator** (E1B10). RACES does not have its own band; you use your normal class privileges.	E1B10

Dessert 13 Question Bank (E1B)

E1B01. Which of the following constitutes a spurious emission?
A. An amateur station transmission made without the proper call sign identification.
B. A signal transmitted to prevent its detection by any station other than the intended recipient.
C. Any transmitted signal that unintentionally interferes with another licensed radio station and whose levels exceed 40 dB below the fundamental power level.
D. An emission outside the signal's necessary bandwidth that can be reduced or eliminated without affecting the information transmitted.

E1B02. Which of the following is an acceptable bandwidth for digital voice or slow-scan TV transmissions made on the HF amateur bands?
A. 3 kHz.
B. 10 kHz.
C. 15 kHz.
D. 20 kHz.

E1B03. Within what distance must an amateur station protect an FCC monitoring facility from harmful interference?
A. 1 mile.
B. 3 miles.
C. 10 miles.
D. 30 miles.

E1B04. What must the control operator of a repeater operating in the 70-centimeter band do if a radiolocation system experiences interference from that repeater?
A. Reduce the repeater antenna HAAT (Height Above Average Terrain).
B. File an FAA NOTAM (Notice to Air Missions) with the repeater system's ERP, call sign, and six-character grid locator.
C. Cease operation or make changes to the repeater that mitigate the interference.
D. All these choices are correct.

E1B05. What is the National Radio Quiet Zone?
A. An area surrounding the FCC monitoring station in Laurel, Maryland.
B. An area in New Mexico surrounding the White Sands Test Area.
C. An area surrounding the National Radio Astronomy Observatory.
D. An area in Florida surrounding Cape Canaveral.

E1B06. Which of the following additional rules apply if you are erecting an amateur station antenna structure at a site at or near a public use airport?
A. You may have to notify the Federal Aviation Administration and register it with the FCC as required by Part 17 of the FCC rules.
B. You may have to enter the height above ground in meters, and the latitude and longitude in degrees, minutes, and seconds on the FAA website.
C. You must file an Environmental Impact Statement with the EPA before construction begins.
D. You must obtain a construction permit from the airport zoning authority per Part 119 of the FAA regulations.

E1B07. To what type of regulations does PRB-1 apply?
A. Homeowners associations.
B. FAA tower height limits.
C. State and local zoning.
D. Use of wireless devices in vehicles.

E1B08. What limitations may the FCC place on an amateur station if its signal causes interference to domestic broadcast reception, assuming that the receivers involved are of good engineering design?
A. The amateur station must cease operation.
B. The amateur station must cease operation on all frequencies below 30 MHz.
C. The amateur station must cease operation on all frequencies above 30 MHz.
D. The amateur station must avoid transmitting during certain hours on frequencies that cause the interference.

E1B09. Which amateur stations may be operated under RACES rules?
A. Only those club stations licensed to Amateur Extra class operators.
B. Any FCC-licensed amateur station except a Technician class.
C. Any FCC-licensed amateur station certified by the responsible civil defense organization for the area served.
D. Only stations meeting the FCC Part 97 technical standards for operation during an emergency.

E1B10. What frequencies are authorized to an amateur station operating under RACES rules?
A. All amateur service frequencies authorized to the control operator.
B. Specific segments in the amateur service MF, HF, VHF, and UHF bands.
C. Specific local government channels.
D. All these choices are correct.

E1B11. What does PRB-1 require of state and local regulations affecting amateur radio antenna size and structures?
A. No limitations may be placed on antenna size or placement.
B. Reasonable accommodations of amateur radio must be made.
C. Such structures must be permitted when use for emergency communications can be demonstrated.
D. Such structures must be permitted if certified by a registered professional engineer.

Answer Key

E1B01: D	E1B02: A	E1B03: A	E1B04: C
E1B05: C	E1B06: A	E1B07: C	E1B08: D
E1B09: C	E1B10: A	E1B11: B	